

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283267

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Shafer; Glen

MODULAR CHEMICAL DISPENSER AND PUMP FOR SAME

Abstract

A chemical dispenser includes a housing, a controller disposed in the housing for operating the chemical dispenser, at least one module bay in the housing and at least one module selectively coupled to the at least one module bay and operatively coupled to the controller for operation with the chemical dispenser. The at least one module may be selected from a plurality of modules each capable of being coupled to the at least one module bay and operating under the control of the controller. A low-maintenance piston pump module for use with the chemical dispenser is also disclosed.

Inventors: Shafer; Glen (Amelia, OH)

Applicant: Delaware Capital Formation, Inc. (Wilmington, DE)

Family ID: 71579667

Appl. No.: 19/094191

Filed: March 28, 2025

Related U.S. Application Data

parent US continuation 17619294 20211215 parent-grant-document US 12264428 US continuation PCT/US2020/039049 20200623 child US 19094191
us-provisional-application US 62865461 20190624

Publication Classification

Int. Cl.: D06F39/02 (20060101); A47L15/44 (20060101); A47L15/46 (20060101); D06F105/42 (20200101); D06F105/60 (20200101); F04B41/06 (20060101); F04B53/10 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/619,294, filed on Dec. 15, 2021, which is a National Stage Application under 35 USC § 371 and claims the benefit of International Patent Application No. PCT/US2020/039049, filed on Jun. 23, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/865,461, filed on Jun. 24, 2019, which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] This invention generally relates to an improved chemical dispenser for a chemical dispensing system, and more particularly to a chemical dispenser having a modular design and an improved pump for the modular chemical dispenser.

BACKGROUND

[0003] The dispensing of liquid chemical products from one or more chemical receptacles is a common requirement of many industries, such as the laundry, textile, warewash, healthcare, and food processing industries. In an industrial laundry facility, for example, one of several operating washing machines will require, from time to time, aqueous solutions containing quantities of alkaloid, detergent, bleach, starch, softener and/or sour. By way of further example, in industrial warewash applications, washing machines will require quantities of detergent, rinse aid, and/or sanitizer. Increasingly, such industries have turned to automated methods and systems for dispensing chemical products.

[0004] Contemporary automatic chemical dispensing systems used in the commercial washing industry typically rely on pumps to deliver liquid chemical products from bulk storage containers. Generally, these pumps deliver raw product to a washing machine either directly or via a flush manifold, where the product is mixed with a diluent, such as water, that delivers the chemical product to the machine. A typical chemical dispensing system used to supply a washing machine will include a controller that is coupled to one or more peristaltic pumps in a dispenser by a plurality of dedicated signal lines. The controller will also typically be coupled to a washing machine interface by another plurality of dedicated signal lines, so that the controller is provided with signals indicating the operational state of the machine. In operation, the machine interface transforms high voltage trigger signals generated by the washing machine into lower voltage signals suitable for the controller, and transmits these low voltage trigger signals to the controller over the set of dedicated signal lines, which are typically in the form of a multi-conductor cable. In response to these individual trigger signals, the controller will individually activate one or more of the pumps in the dispenser over another set of dedicated lines so that the pumps dispense a desired amount of a chemical product into the washing machine or into the flush manifold, where the chemicals are then mixed with a diluent before being delivered to the machine.

[0005] Chemical dispensing systems employed with commercial washing machines typically utilize peristaltic pumps to minimize both operator and system component contact with the chemical products, which are often corrosive and toxic. Peristaltic pumps of this type include a flexible tube (or squeeze tube) and a rotor with one or more rollers located in a pump chamber. The one or more rollers compress a section of the squeeze tube against a wall of a pump chamber, pinching off the section of squeeze tube. When the rotor is rotated, the location of the pinched

section of the squeeze tube moves along the length of the tube, thereby forcing, or pumping, fluid through the tube. While peristaltic pumps operate for their intended purpose, there are some drawbacks to current chemical dispensers employing peristaltic pumps.

[0006] By way of example, chemical dispensers with peristaltic pumps generally require regular maintenance to ensure proper operation of the chemical dispensing system. In this regard, the squeeze tubes used in such pumps are subject to wear over time from the repeated compression and pulling from the rollers, which causes the volume of chemical pumped by the dispenser to vary over time. Worn out squeeze tubes must be regularly replaced to prevent tube failure. Moreover, squeeze tube replacement can be a cumbersome endeavor, as chemical product often leaks from the feed lines when the seal is broken between the squeeze tube and feeder tubes. In addition to causing a loss of product and undesirably exposing workers to potentially hazardous chemicals, the spilled product may also contaminate the surfaces of the squeeze tube and pump chamber. If the chemical product is not sufficiently cleaned from these surfaces, the resulting sticky residue can cause the roller to pull the squeeze tube through the pump chamber so that the tube becomes damaged or tangled, resulting in pump failure and further potential product spills. In addition, because the controller cannot determine that the pump is not dispensing the correct amount of product, any processed wash loads that rely on the failed pump will have to be re-processed. Further, because the timing of the pump failure may be difficult to determine, multiple wash loads may have to be reprocessed.

[0007] In addition to the above, current chemical dispensers typically have the pumps integrated into the chemical dispenser housing. Thus, while different types of pumps may be available and preferred, depending on the chemical product being dispensed, the use of alternative pumps require a wholesale replacement of the chemical dispenser. More particularly, the chemical dispenser may have to be specifically designed to include different types of pumps for different applications and chemical products. Such an approach to designing an optimal chemical dispensing system is cost prohibitive.

[0008] Therefore, there is a need for a chemical dispensing system having an improved chemical dispenser that allows different types of pumps to be used for different applications in an easy and cost-effective manner. There is also a need for an improved pump for the chemical dispenser that operates accurately and requires less maintenance.

SUMMARY

[0009] The present invention overcomes the foregoing and other shortcomings and drawbacks of chemical dispensing systems, chemical dispensers, and modular pumps. While the invention will be described in connection with certain embodiments, it will be understood that the invention is not limited to these embodiments. On the contrary, the invention includes all alternatives, modifications and equivalents as may be included within the spirit and scope of the present invention.

[0010] According to one aspect of the present invention, there is a chemical dispenser including a housing, a controller disposed in the housing for operating the chemical dispenser, at least one module bay in the housing, and at least one module selectively coupled to the at least one module bay and operatively coupled to the controller for operation with the chemical dispenser. The at least one module is selected from a plurality of modules each capable of being coupled to the at least one module bay and operating under the control of the controller. In one embodiment, the housing includes a plurality of module bays, each module bay is configured to receive a respective module selected from the plurality of modules.

[0011] In one embodiment, at least one of the plurality of modules is a pump. For example, more than one of the plurality of modules may be pumps and can include one or more of peristaltic pumps, diaphragm pumps, dual-piston pumps, and/or double-ended piston pumps. In one embodiment, at least one of the plurality of modules is an alarm. For example, more than one of the plurality of modules are alarms and can include visual alarms and/or audio alarms. In one

embodiment, at least one of the plurality of modules is a valve. For example, more than one of the plurality of modules are valves and can include a solenoid valve.

[0012] According to another aspect, a chemical dispensing system comprises the chemical dispenser of any of the embodiments.

[0013] According to another aspect, a washing arrangement comprises a washing machine and a chemical dispensing system according to one aspect operatively coupled to the washing machine.

[0014] According to yet another aspect, a pump module for a modular chemical dispensing system comprises a module housing, a piston assembly, a drive assembly, and a valve assembly.

[0015] In one embodiment, the piston assembly comprises a piston housing defining at least two piston cylinders. At least two pistons each define a base and a piston head for positioning in respective piston cylinders. The base of the pistons is operatively coupled to the drive assembly for reciprocating the pistons relative to the piston cylinders.

[0016] In one embodiment, the piston housing includes at least one guide channel, and each piston includes at least one guide rod. The at least one guide rod is configured to be received in a respective guide channel for guiding the movement of the pistons.

[0017] In one embodiment, the drive assembly comprises a motor having a drive shaft coupled to the module housing and a gear arrangement operatively coupled to the motor and further operatively coupled to the piston assembly.

[0018] In one embodiment, the gear arrangement comprises a primary drive gear coupled to the drive shaft of the motor and a pair of secondary drive gears configured to be driven by the primary drive gear. In one embodiment, each of the secondary drive gears includes a pin eccentrically positioned relative to a rotational axis of the secondary drive gears. The pins are configured to be received within a slot in the base of the pistons for moving the pistons.

[0019] In one embodiment, the valve assembly comprises a valve housing, a pair of valves, and a product manifold. In one embodiment, the valve housing comprises a pair of valve heads. Each valve head includes a valve recess. Each valve recess includes an inlet port, an outlet port, and a valve seat. The valve seat is configured to receive one of the pair of valves. The inlet and outlet ports of each valve head are in communication with a respective one of the piston chambers. In one embodiment, the inlet port includes at least one flow aperture and a valve post. In one embodiment, the inlet port includes a pair of flow apertures with the valve post disposed therebetween.

[0020] In one embodiment, the outlet port includes an annular valve seat.

[0021] In one embodiment, the product manifold comprises an inlet channel and an outlet channel. The inlet channel is in communication with the inlet ports of each of the valve heads, and the outlet channel is in communication with the outlet ports of each of the valve heads.

[0022] In one embodiment, the piston assembly comprises a piston housing defining at least two piston cylinders, and a piston having a sliding yoke and two piston heads extending in opposing directions from the sliding yoke. Each one of the piston heads is received in a respective piston cylinder. The sliding yoke is operatively coupled to the drive assembly for reciprocating the piston relative to the two piston cylinders.

[0023] In one embodiment, one cycle of the piston in the piston assembly is configured to produce two exhaust and two intake cycles.

[0024] In one embodiment, the piston heads share a common longitudinal axis.

[0025] In one embodiment, the opposing piston heads are of different lengths.

[0026] In one embodiment, the piston heads are hollow and are open to the respective piston cylinder.

[0027] In one embodiment, the sliding yoke defines an elliptical slot and the drive assembly is movably coupled to the sliding yoke by the elliptical slot.

[0028] In one embodiment, the piston housing further defines an opening in a yoke cavity. The yoke cavity receives the sliding yoke, and the drive assembly engages the piston assembly through the opening.

[0029] In one embodiment, the piston assembly further comprises a first cylinder head secured in the piston housing and at least partially defining a portion of one piston cylinder and a second cylinder head secured in the piston housing and at least partially defining a portion of the other piston cylinder. In one embodiment, each of the first cylinder head and the second cylinder head are in fluid communication with one piston.

[0030] In one embodiment, the first cylinder head is in fluid communication with a first piston and the second cylinder head is in fluid communication with a second piston. The first piston and second piston are different pistons.

[0031] In one embodiment, the valve assembly comprises an inlet valve housing including an inlet valve in fluid communication with at least one cylinder and an outlet valve housing including an outlet valve in fluid communication with the at least one cylinder. In one embodiment, each valve is a duckbill valve.

[0032] In one embodiment, the piston housing defines two cylinders and the piston assembly further comprises a first cylinder head secured in the piston housing and at least partially defining a portion of one piston cylinder. The piston housing further comprises a second cylinder head secured in the piston housing and at least partially defining a portion of the other piston cylinder. And, the valve assembly comprises a first inlet valve housing including a first inlet valve coupled to the first cylinder head and a first outlet valve housing including a first outlet valve coupled to the first cylinder head. A second inlet valve housing includes a second inlet valve coupled to the second cylinder head, and a second outlet valve housing includes a second outlet valve coupled to the second cylinder head. Each of first inlet valve, the first outlet valve, the second inlet valve, and the second outlet valve is a duckbill valve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with a general description of the invention given above, and the detailed description given below, serve to explain the invention.

[0034] FIG. 1 is an illustration of an exemplary chemical dispensing system having a chemical dispenser in accordance with an embodiment of the present invention;

[0035] FIG. 1A is another illustration of an exemplary chemical dispensing system having a chemical dispenser in accordance with an embodiment of the present invention;

[0036] FIG. 2 is a perspective view of a chemical dispenser in accordance with an embodiment of the present invention;

[0037] FIG. 3 is a partially disassembled perspective view of the chemical dispenser shown in FIG. 2;

[0038] FIG. 4 is a perspective view of a chemical dispenser in accordance with another embodiment of the present invention;

[0039] FIG. 5 is a disassembled perspective view of a dual-piston pump module in accordance with an embodiment of the present invention;

[0040] FIG. 5A is a partially disassembled perspective view of a valve arrangement for the dual-piston pump module shown in FIG. 5;

[0041] FIG. 5B is a partially disassembled perspective view of a piston assembly for the dual-piston pump module shown in FIG. 5;

[0042] FIG. 5C is a partial perspective view of a drive assembly for the dual-piston pump module shown in FIG. 5;

[0043] FIG. 6A is a cross-sectional view of the dual-piston pump module illustrating the inflow of chemical product to the pump;

[0044] FIG. 6B is another cross-sectional view of the dual-piston pump module illustrating the inflow of chemical product to the pump;

[0045] FIG. 6C is an enlarged partial view of the valve arrangement during the inflow of chemical product to the pump;

[0046] FIG. 6D is another enlarged partial view of the valve arrangement during the inflow of chemical product to the pump;

[0047] FIG. 7A is a cross-sectional view of the dual-piston pump module illustrating the outflow of chemical product from the pump;

[0048] FIG. 7B is another cross-sectional view of the dual-piston pump module illustrating the outflow of chemical product from the pump;

[0049] FIG. 7C is an enlarged partial view of the valve arrangement during the outflow of chemical product from the pump; and

[0050] FIG. 7D is another enlarged partial view of the valve arrangement during the outflow of chemical product from the pump;

[0051] FIG. 8 is a perspective view of a double-ended piston pump module in accordance with an embodiment of the present invention;

[0052] FIG. 9 is a disassembled perspective view of the pump module shown in FIG. 8;

[0053] FIG. 10 is a perspective view of a piston of the pump module shown in FIG. 9;

[0054] FIG. 11 is a cross-sectional view of the piston of FIG. 10 taken along section line 11-11;

[0055] FIG. 12 is a cross-sectional view of the pump module shown in FIG. 8 illustrating lateral movement of the piston;

[0056] FIG. 12A is an enlarged cross-sectional view of the pump module of FIG. 12 illustrating fluid movement from one cylinder due to piston motion;

[0057] FIG. 12B is an enlarged cross-sectional view of the pump module of FIG. 12 illustrating fluid movement into the other cylinder due to the same piston motion;

[0058] FIG. 13 is a cross-sectional view of the pump module shown in FIG. 8 illustrating lateral movement of the piston;

[0059] FIG. 13A is an enlarged cross-sectional view of the pump module of FIG. 13 illustrating fluid movement into due to piston motion;

[0060] FIG. 13B is an enlarged cross-sectional view of the pump module of FIG. 13 illustrating fluid movement from the other cylinder due to the same piston motion;

[0061] FIG. 14 is a perspective view of a dual-piston pump module in accordance with an embodiment of the present invention;

[0062] FIG. 15 is a disassembled perspective view of the pump module shown in FIG. 14;

[0063] FIG. 15A is a partially disassembled perspective view of a valve assembly for the piston pump module shown in FIG. 14;

[0064] FIG. 15B is a perspective view of a piston of the pump module shown in FIG. 14;

[0065] FIG. 15C is a cross-sectional view of the piston of FIG. 15B taken along section line 15C-15C;

[0066] FIG. 16A is a cross-sectional view of the dual-piston pump module illustrating the outflow of chemical product to from pump;

[0067] FIG. 16B is another cross-sectional view of the dual-piston pump module illustrating the inflow of chemical product to the pump;

[0068] FIG. 17A is an enlarged partial view of the valve assembly during the outflow of chemical product from the pump;

[0069] FIG. 17B is another enlarged partial view of the valve arrangement during the outflow of chemical product from the pump;

[0070] FIG. 18A is an enlarged partial view of the valve assembly during the inflow of chemical product to the pump;

[0071] FIG. 18B is another enlarged partial view of the valve arrangement during the inflow of

chemical product to the pump;

[0072] FIG. **19A** is a cross-sectional view of the dual-piston pump module illustrating the outflow of chemical product to from pump;

[0073] FIG. **19B** is another cross-sectional view of the dual-piston pump module illustrating the inflow of chemical product to the pump;

[0074] FIG. **20A** is an enlarged partial view of the valve assembly during the outflow of chemical product from the pump;

[0075] FIG. **20B** is another enlarged partial view of the valve arrangement during the outflow of chemical product from the pump;

[0076] FIG. **21A** is an enlarged partial view of the valve assembly during the inflow of chemical product to the pump; and

[0077] FIG. **21B** is another enlarged partial view of the valve arrangement during the inflow of chemical product to the pump.

DETAILED DESCRIPTION

[0078] With reference to FIG. **1**, an exemplary chemical dispensing system **10** for use with a washing machine **12**, which may be a laundry machine is illustrated. The chemical dispensing system **10** includes a chemical dispenser **14**, having at least one and preferably a plurality of pumps **16a**, **16b**, one or more chemical reservoirs **18a**, **18b** in fluid communication with respective pumps **16a**, **16b** via input product lines **20a**, **20b**, and a fluid manifold **22** in fluid communication with each of the pumps **16a**, **16b** via output product lines **24a**, **24b**. For laundry applications, there may be as many as eight pumps, reservoirs, and associated product lines. The fluid manifold **22** is in fluid communication with the washing machine **12** via a machine supply line **26** and is in further fluid communication with a diluent source **28** via a diluent supply line **30**. The diluent supply line **30** may include a valve **32** operatively coupled to the chemical dispenser **14** for controlling the flow of diluent through the fluid manifold **22** and to the washing machine **12**. In this regard, the chemical dispenser **14** may include a controller **34** for controlling the chemical dispenser, including, for example, the pumps **16a**, **16b** and the valve **32**. Additional details of the controller **34** are provided in U.S. Application Ser. No. 62/843,777 ("the '777 application"), filed on May 6, 2019 and titled Dispensing System. The disclosure of the '777 application is incorporated by reference herein in its entirety.

[0079] FIG. **1A** illustrates another chemical dispensing system **10a** for use with a washing machine **12a**, which in the illustrated embodiment may be a warewash machine. The chemical dispensing system **10a** includes a chemical dispenser **14**, having at least one and preferably a plurality of pumps **16a**, **16b**, one or more chemical reservoirs **18a**, **18b** in fluid communication with respective pumps **16a**, **16b** via input product lines **20a**, **20b**, and output product lines **24a**, **24b** in communication with the washing machine **12a**. In this application, for example, the fluid manifold **22** may be omitted and the pumps **16a**, **16b** may be directly coupled to the washing machine **12a**. For warewash applications, there may be as many as three pumps, reservoirs, and associated product lines. It should be recognized that aspects of the present invention are not limited to laundry and warewash applications but may apply to a host of other industries including the textile, healthcare, and food processing industries. Additionally, aspects of the invention are not limited to any particular number of pumps, reservoirs, product lines, etc., which may be based on the particular application.

[0080] FIG. **2** illustrates a chemical dispenser **14** in accordance with an exemplary embodiment of the invention. The chemical dispenser **14** includes an outer housing **40** for holding the one or more pumps **16a**, **16b** and a controller **34**. In one embodiment, the housing **40** may be generally rectangular in shape and include a front panel **42**, rear panel **44**, top panel **46**, bottom panel **48**, and side panels **50**, **52** that collectively define a housing interior **54**. It should be recognized, however, that the housing **40** is not limited to this shape as other housing shapes and configurations are possible within the scope of the invention. The housing **40** may be formed from a suitable material,

such as a strong engineering plastic, through an injection molding process, for example. Other materials and forming processes are also possible.

[0081] The chemical dispenser **14** may be configured to be mounted to a wall or stand at an industrial facility or the like in relatively close proximity to the washing machine **12**. In this regard, the rear panel **44** may include various fasteners or features that facilitate the mounting of the chemical dispenser within the facility. The front panel **42** of the chemical dispenser **14** generally includes a controller section **60** and a module section **62**. In one embodiment, the controller section **60** occupies an upper portion of the front panel **42** of the chemical dispenser **14** and the module section **62** occupies a lower portion of the front panel **42** of the chemical dispenser **14**. The invention, however, is not limited to such an arrangement as the controller section **60** and the module section **62** may be reversed or alternatively placed side-by-side.

[0082] The controller section **60** includes various features for a user to interact with the controller **34** and/or observe performance features of the chemical dispenser **14**. By way of example, the controller section **60** may include various buttons, such as standby buttons, prime buttons, etc., and/or various indicators, such as dispenser status indicators (e.g., light-emitting diodes), pump status indicators, etc. The controller section **60** may further include a user input interface (e.g., touchscreen) and/or user output interface. Additional details of the controller section **60** may be found in the '777 application.

[0083] In accordance with an aspect of the invention, the chemical dispenser **14** is configured to be modular and capable of receiving a variety of different types of modules in the housing **40** in a plug-and-play manner. In this regard and as illustrated in FIGS. 2 and 3, the module section **62** is configured to include a plurality of module bays **64a**, **64b** each configured to receive a module **66a**, **66b** for use with the chemical dispenser **14**. While two module bays **64a**, **64b** and corresponding modules **66a**, **66b** are shown with chemical dispenser **14**, it should be recognized that the module section **62** of the chemical dispenser **14** may include more or fewer bays and modules. FIG. 4, for example, illustrates three module bays **64a**, **64b**, **64c** and corresponding modules **66a**, **66b**, **66c**. Thus, the chemical dispenser **14** may include most any desired number of module bays **64** and modules **66** to meet the needs of a particular application.

[0084] As illustrated in FIG. 3, each module bay **64a**, **64b** includes a generally rectangular support surface **68** having an aperture **70** open to the interior **54** of the housing **40**. The support surface **68** may also include one or more fastening elements for securing a module **66** to a respective module bay **64**. For example, in an exemplary embodiment, the support surface **68** may include one or more threaded bores **72** configured to receive a screw (not shown) for securing a module **66** to a module bay **64**. The invention is not limited to such fastening elements. For example, other types of fasteners may be used to secure a module **66** to a module bay **64**, including various clamps, clips, latches, magnets, etc. In any event, the module **66** may be easily and selectively coupled and decoupled from the modular bays **64**.

[0085] As further illustrated in FIGS. 2 and 3, each module **66** includes a generally rectangular face plate **74** configured to engage with the support surface **68** of the module bays **64** when the modules **66** are coupled to the chemical dispenser **14**. In this regard, the modules **66** may include one or more fastening elements (not shown) for securing the modules **66** to a respective module bay **64**. The modules **66** may each have a substantially similar size and be configured to mount to any of the module bays **64** on the chemical dispenser **14**. Moreover, the modules **66** may provide a variety of functions to the chemical dispenser **14**. For example, in one embodiment a module **66** may be configured as a pump for the chemical dispenser **14**. In another embodiment, the module **66** may be configured as an alarm for the chemical dispenser **14**. In yet another embodiment, the module **66** may be configured as a valve for the chemical dispenser **14**. Thus, the modules **66** may be different from each other but yet be configured to be mounted to any of the module bays **64** in the dispenser housing **40**. Furthermore, each of the module bays **64** may include an interface, such as a wire harness (not shown), for operatively coupling the modules **66** to the controller **34**, thereby allowing

the controller **34** to control operation of the modules **66** coupled to the module bays **64**. This type of modularity and plug-and-play capability provides designers, manufacturers, and consumers of chemical dispensing systems a wider range of options when designing a laundry or wash-ware application, for example.

[0086] As noted above, the module **66** may take the form of a pump **16**. In accordance with an aspect of the invention, the pump **16** may be one of several designs each configured to be mounted to a module bay **64** of the chemical dispenser **14**. By way of example and without limitation, the module **66** may be configured as a peristaltic pump. Alternatively, the module **66** may be configured as a diaphragm pump. Still further, and as discussed in more detail below, the module **66** may be configured as a dual-piston pump or double-ended piston pump. Thus, depending on the particular application and the desire of the consumer, different types of pumps **16** may be coupled to the chemical dispenser **14** in an interchangeable manner and without any difficulty. As illustrated in FIGS. **2** and **3**, each of the pumps **16** includes an inlet **76** configured to be coupled to an input product line **20** from a chemical reservoir **18**, and an outlet **78** configured to be coupled to an output product line **24** connected to the fluid manifold **22**. The pumps **16** associated with the chemical dispenser **14** may all be the same type of pump **16** or may be different from each other. For example, a peristaltic pump may be positioned in one of the module bays **64** while a dual-piston pump or double-ended piston pump may be positioned in another module bay **64**. Thus, a great variety of pumps and arrangements in the chemical dispenser **14** are possible in embodiments of the present invention.

[0087] As illustrated in FIG. **4**, a module **66c** may take the form of an alarm **80** configured to notify a user when an error condition of the chemical dispenser **14** is detected by the controller **34**. In one embodiment, the alarm **80** may be a visual alarm having, for example, different colored lights that indicate the operation of the chemical dispenser **14**. By way of example, when the chemical dispenser **14** is operating normally, the alarm **80** may illuminate as a green light. When a non-emergency error condition exists in the chemical dispenser **14**, the alarm **80** may illuminate as a yellow light indicating that action should be taken in the near future. Furthermore, when an error condition is detected that requires immediate attention, the alarm **80** may illuminate as a red light. The invention is not limited to this arrangement of lights and it should be recognized that a module **66** may include a different type of visual alarm.

[0088] In an alternative embodiment, the alarm **80** may be configured as an audio alarm having, for example, different sounds or frequency of sounds that indicate the operation of the chemical dispenser. Thus, by way of example, when the chemical dispenser **14** is operating normally, the alarm **80** may project a first sound at a first frequency (e.g., low frequency). When a non-emergency error condition exists in the chemical dispenser **14**, the alarm **80** may project a second sound at a second frequency (slightly higher frequency) indicating that action should be taken in the near future. Furthermore, when an error condition is detected that requires immediate attention, the alarm **80** may project a third sound at a third frequency (e.g., high frequency). The invention is not limited to this arrangement of sounds/frequency and it should be recognized that a module **66** may include a different type of audio alarm.

[0089] In yet a further embodiment, a module **66** may be configured as a valve (not shown), such as, for example, a solenoid valve. By way of example, the valve module **66** of this embodiment may take the place of valve **32** (FIG. **1**) such that the diluent source **28** is now in fluid communication with an inlet of a module **66** of the chemical dispenser **14** and the fluid manifold **22** is in fluid communication with an outlet of the module. Thus, a module **66** of the chemical dispenser **14** controls the flow of diluent through the chemical dispensing system **10**.

[0090] While the modules **66** of the chemical dispenser **14** have been described herein as pumps, alarms, and valves, it should be recognized that modules providing other functions may be possible and within the scope of the present invention. By way of example, other functionalities that may be performed by one or more modules **66** include various types of out-of-product indicators, such as

optical or other types of indicators, and/or proof of delivery indicators that confirm the delivery and/or amount of chemical product dispensed to the washing machine.

[0091] The modular design of the chemical dispenser **14** provides a number of advantages. As an initial matter, the chemical dispenser **14** provides a versatile design that allows designers, manufacturers and customers to configure a dispenser that meets their specific needs. The plug-and-play feature of the modules **66** allows the chemical dispenser **14** to be easily configured or reconfigured for a particular application. Additionally, performing maintenance on the chemical dispenser **14** has been greatly enhanced. For example, should a pump **16** of the chemical dispenser **14** stop working properly, the malfunctioning pump may be removed from the housing **40** and replaced with a new or refurbished pump in a quick and relatively easy repair procedure. In short, the chemical dispenser **14** is versatile and may be configured to meet the needs in a wide range of applications and configurations. Moreover, the interchangeability of the modules improves maintenance/repairs and reduces outages of the chemical dispensing system **10**.

[0092] FIGS. 5-7D illustrate an improved pump module **90** in accordance with an embodiment of the invention. The pump module **90** may be just one of the types of modules **66** used in chemical dispenser **14** described above. In accordance with an aspect of the invention, the pump module **90** may be configured as a dual-piston pump capable of relatively constant fluid flow over fairly short cycle times. The dual-piston pump module **90** is also configured to be low maintenance and capable of very long run times before any maintenance operations are necessary to ensure the accurate dispensing of chemical product from the chemical dispenser **14**. This further reduces the maintenance costs and down time for the chemical dispensing system **10**.

[0093] A disassembled dual-piston pump module **90** in accordance with an embodiment of the invention is illustrated in FIG. 5 and broadly includes a module housing **92**, a piston assembly **94**, a drive assembly **96**, and a valve assembly **98**. The module housing **92** includes a front housing portion **100** and a rear housing portion **102** which fit together to form the module housing **92** with an interior **104** for housing the components of the pump. The rear housing portion **102** includes a generally planar wall **106**, a U-shaped support or frame **108** extending from an inner surface of the wall **106**, a pair of spindles **110** extending from the wall **106** within the U-shaped frame **108**, and a pair of support posts **112** extending from the wall **106** above and outboard of the U-shaped frame **108**. The rear housing portion **102** further includes a drive aperture **114** in the wall **106** centrally located above and between the spindles **110** and a pair of slots **116**, the purpose of which will be described below, at a lower end of the rear housing portion **102**. The front housing portion **100** generally defines a cavity **118** and effectively operates as a cover for the internal components of the pump module **90**. The front and rear housing portions **100**, **102** may be coupled together by fasteners, such as screws, which are received in threaded bores in the rear housing portion **102**. For example, the ends of the posts **112** may include threaded bores and the U-shaped frame **108** may include a threaded bore. Other fastening arrangements are possible, however. Additionally, the front and rear housing portions **100**, **102** may be made (e.g., molded) from suitable engineering plastics.

[0094] As illustrated in FIGS. 5 and 5B, the piston assembly **94** includes a piston chamber housing **122** defining a pair of piston chambers **124** and a pair of pistons **126** each configured to be received within a respective piston chamber **124** of the piston chamber housing **122**. The piston chambers **124** are defined by respective generally cylindrical walls or piston cylinders **128** that are open at both an upper end and lower end thereof. The piston chamber housing **122** further includes a pair of guide channels **130** on opposing sides of each of the cylindrical walls **128** that define the piston chambers **124**. The purpose of the guide channels **130** is explained in more detail below. The lateral ends of the piston chamber housing **122** further include a pair of support tubes **131** for securing the piston chamber housing **122** to the pump module **90**, and more particularly to the rear housing portion **102** of the module housing **92**. In this regard, the piston chamber housing **122** is sized to fit generally between the posts **112** such that the support tubes **131** are configured to be slidably received over the posts **112**.

[0095] With reference to FIG. 5B, each of the pistons **126** include a generally circular base **132** and an elongate stem **134** extending from the base **132** and terminating in a piston head **136**. The base **132** includes a generally oval or elliptical slot **138** configured to receive a portion of the drive assembly **96** for moving the pistons **126** relative to the piston chambers **124**, as will be discussed in more detail below. The piston heads **136** are sized to be slidably received within the piston chambers **124** of the piston chamber housing **122**. In this regard, the piston heads **136** may include one or more seals (e.g., O-rings) that form a substantially fluid tight interface between the piston heads **136** and the cylindrical walls **128** during operation of the pump module **90**. In addition, the pistons **126** may include a pair of guide rods **140** extending from the base **132** on opposed sides of the stem **134** and configured to be received within the guide channels **130** in the piston chamber housing **122** during operation. The interaction between the guide rods **140** on the pistons **126** and the guide channels **130** in the piston chamber housing **122** maintains the movement of the pistons **126** in a single direction, e.g., in a substantially vertical direction.

[0096] As illustrated in FIGS. 5 and 5C, the drive assembly **96** includes a motor **146** and a gear arrangement **148** operatively coupled to the motor **146** and to the piston assembly **94** for reciprocating the pistons **126** within the piston chambers **124**. As illustrated in FIG. 5, the motor **146** is configured to be coupled to the module housing **92** and includes a rotatable drive shaft **150** extending from the motor **146** and into the interior **104** of the module housing **92**. In this regard, the wall **106** of the rear housing portion **102** includes one or more bores configured to receive fasteners (e.g., screws) that secure the motor **146** to the wall **106** of the rear housing portion **102**. When so secured, the drive shaft **150** extends through the drive aperture **114** in the wall **106** of the rear housing portion **102**. As is shown in FIG. 5C, the gear arrangement **148** includes a primary drive gear **152** and a pair of secondary drive gears **154**. The primary drive gear **152** is received on the drive shaft **150** of the motor **146** and is rotatably driven by the motor **146**. The secondary drive gears **154** are each received on a respective spindle **110** and are configured to mesh with the primary drive gear **152** such that the secondary drive gears **154** are rotatably driven by the primary drive gear **152** with activation of the motor **146**. In one embodiment, the ratio between the primary and second gears may be 1:1 such that a single rotation of the primary gear results in a single rotation of the secondary gears **154**. The invention is not limited to this ratio, however, as other gear ratios are possible depending on the particular application, for example.

[0097] The secondary drive gears **154** each include an eccentrically located pin **156** extending from a face of the secondary drive gears **154**. For example, the pins **156** may be located adjacent an outer portion of the drive gears **154** such that the pins **156** rotate about the central axis of the secondary drive gears **154**. As illustrated in FIGS. 6A, 6B, 7A and 7B, each pin **156** is configured to be received within a respective elliptical slot **138** in the base **132** of respective pistons **126**. As the secondary drive gears **154** rotate, the eccentrically located pins **156** slide within the slots **138** in the pistons **126** (e.g., side-to-side) and also move the pistons **126** vertically within and relative to the piston chambers **124** of the piston-chamber housing **122**. The secondary drive gears **154** and associated pins **156** may be arranged such that when one of the pistons **126** is positioned at top dead center relative to its piston chamber **124**, the other piston **126** is positioned at bottom dead center relative to its piston chamber **124** (i.e., the pistons **126** are at opposite ends of their respective strokes). Thus, when the motor **146** is energized, the primary drive gear **152** drives the secondary drive gears **154**, which in turn cause reciprocating movement of the pistons **126** within their respective piston chambers **124**. The use of a dual-piston arrangement as a pump, however, involves the coordinated use of a valve arrangement, to which we now turn.

[0098] As illustrated in FIGS. 5 and 5A, the valve assembly **98** includes a valve housing **162**, a pair of valves **164**, and a product manifold **166**. As illustrated in more detail in FIG. 5A, the valve housing **162** includes a pair of valve heads **168** configured to be positioned above the piston chambers **124** of the piston chamber housing **122**. As illustrated in FIGS. 6A, 6B, 7A, 7B, each of the valve heads **168** include a bore **170** configured to receive a portion of the piston chamber

housing **122** therein. Moreover, each of the valve heads **168** include a generally elliptical valve recess manifold **171** that defines an inlet port **172**, and outlet port **174** and an outer valve seat **176** positioned about the inlet and outlet ports **172**, **174** for receiving a valve **164**. The inlet and outlet ports **172**, **174** of each valve head **168** are in communication with a respective bore **170**. The inlet port **172** includes at least one and preferably two flow apertures **178** therein and a valve stem or post **180** positioned between the two flow apertures **178**. The outlet port **174** includes an annular valve seat **182** positioned therein and defining an aperture in communication with a respective bore **170**.

[0099] The lateral ends of the valve housing **162** further include a support tube **184** for securing the valve housing **162** to the pump module **90**, and more particularly to the rear housing portion **102** of the module housing **92**. In this regard, the valve housing **162** is sized such that support tubes **184** are configured to be slidably received over the posts **112**. More particularly, as illustrated in FIGS. **6A**, **6B**, **7A**, **7B** when the valve housing **162** and the piston chamber housing **122** are coupled together, the upper ends of the cylindrical walls **128** that define the piston chambers **124** are received in the bores **170** of the valve heads **168** and the support tube **184** of the valve housing **162** fits between and aligns with the support tubes **131** of the piston chamber housing **122** such that the combined assembly may be slidably received over the posts **112** of the module housing **92**. When assembled, the inlet and outlet ports **172**, **174** of each valve head **168** are in communication with a respective piston chamber **124** of the piston assembly **94**. Thus, each piston chamber **124** has associated therewith an inlet port **172** for allowing chemical product into the piston chamber **124** and an outlet port **174** for allowing chemical product to be expelled from the piston chamber **124**.

[0100] As illustrated in FIGS. **5** and **5A**, each of the valves **164** is generally elliptical in shape to correspond to the elliptical shape of the valve seat **176** in the valve recess **171** in the valve heads **168**. Each valve **164** includes a pair of confronting C-shaped cutouts **186** that generally define a pair of generally circular valve flaps **188**, the purpose of which will be described below. When the valves **164** are positioned in the valve seats **176** of the valve housing **162**, one of the valve flaps **188** engages against the annular valve seat **182** in the outlet ports **174**, and the other valve flap **188** engages against the top of the valve post **180** in the inlet ports **172**. This may be envisioned, for example, by moving the valves **164** illustrated in FIG. **5A** down into their respective valve seats **176** in the valve housing **162**. The valves **164** may be made from a suitable elastomeric material that provides some flexing of the material under fluid pressure. For example, the valves **164** may be made from various elastomeric materials, such as fluoroelastomers (e.g., Viton®).

[0101] The product manifold **166** provides for chemical product input to the pump module **90** and chemical product output from the pump module **90** and is configured to be coupled to the valve housing **162**, such as by suitable fasteners. The product manifold **166** includes an inlet channel **190** having a connector **192** at one end and is closed off at the other end **194**, and an outlet channel **196** having a connector **198** at one end and is closed off at the other end **200**. The inlet ports **172** are configured to be in selective communication with the inlet channel **190**, and the outlet ports **174** are configured to be in fluid communication with the outlet channel **196** (e.g., via the valves **164**). The product manifold **166** includes a plurality of ports **202** (see FIGS. **6C**, **6D**, **7C**, and **7D**) that generally overlie and align with the valve recess **171** in the valve heads **168** when the product manifold **166** and valve housing **162** are coupled together. Similar to the above, the product manifold **166** defines a pair of inlet ports **204** and a pair of outlet port **206** corresponding to the inlet and outlet ports **172**, **174** in the valve housing **162**. The configuration of the inlet and outlet ports **204**, **206** in the product manifold **166** are generally opposite to that in the valve housing **162**. Thus, the inlet ports **204** include an annular valve seat **208** and the outlet ports **206** include at least one and preferably two flow apertures **210** with a valve stem or post **212** positioned between the two flow apertures **210**. As further demonstrated in FIGS. **6A**, **6B**, **7A**, **7B** the valve assembly **98** further includes inlet and outlet tubing **214**, **216** extending from their respective connectors **192**, **198** to the inlet and outlet **76**, **78** of the pump module **90**, which may be defined by connectors **218**

that slidably engage with the slots **116** in the module housing **92**.

[0102] To assemble the pump module **90**, the motor **146** may be coupled to the rear housing portion **102** using, for example, suitable fasteners. When so fastened, the drive shaft **150** extends through the drive aperture **114** so as to extend within the interior **104** of the module housing **92**. Next, the gear arrangement **148** may be positioned in the module housing **92**. In this regard, the primary drive gear **152** may be positioned on the drive shaft **150** and the secondary drive gears **154** may be positioned on the spindles **110** so that the teeth of the gears **152**, **154** mesh together. Either prior to the above or subsequent to the above (and separate from the above), the valves **164** may be positioned in their respective valve seats **176** of the valve housing **162** and the product manifold **166** coupled to the valve housing **162** using suitable fasteners. Next, the valve housing/product manifold assembly may be positioned relative to and optionally coupled to the piston chamber housing **122** such that the support tubes **131**, **184** are generally aligned. Next, the pistons **126** may be inserted into their respective piston chambers **124** in the piston chamber housing **122** so that the guide rods **140** engage with their respective guide channels **130**. The pistons **126** may be frictionally held to the piston chamber housing **122**. Next, that entire subassembly may be inserted into the module housing **92** by sliding the aligned support tubes **131**, **184** over the posts **112** and positioning the pistons **126** so that the pins **156** from the secondary drive gears **154** extend into a slot **138** in a respective piston **126**. The inlet and outlet tubing **214**, **216** may then be coupled to connectors **192**, **76** and **198**, **78**, respectively. Lastly, the front housing portion **100** may be coupled to the rear housing portion **102** using suitable fasteners. The pump module **90** is then assembled and ready to be inserted into one of the module bays **64** of the chemical dispenser **14**.

[0103] Operation of the pump module **90**, once coupled to the chemical dispenser **14** and operational within the chemical dispensing system **10**, will now be described. FIGS. **6A-6D** illustrate operation of the pump module **90** as it relates to the inflow of chemical product into the pump, and FIGS. **7A-7D** illustrate operation of the pump module **90** as it relates to the outflow of chemical product from the pump. For purposes of discussion, the initial configuration of the pump module **90** will be with the left piston **126** in the bottom dead position with the piston chamber **124** full of product, and the right piston **126** in the top dead position with the piston chamber **124** fully evacuated. This configuration is shown in FIGS. **6A** and **7A**. Activation of the motor **146** (i.e., under the control of controller **34**) causes the primary drive gear **152** to rotate, which in turn causes both the secondary drive gears **154** to rotate. With rotation of the secondary drive gears **154**, the left piston **126** begins to move upward through a positive pressure stroke and the right piston **126** begins to move downward through a negative pressure stroke (i.e., vacuum).

[0104] Focusing first on the left piston, during the positive pressure stroke of this piston, the positive pressure in the piston chamber **124** causes the valve flap **188** associated with the inlet channel **190** to engage against the annular valve seat **208** such that the valve is closed and fluid cannot pass from the piston chamber **124** to the inlet channel **190**. This valve configuration for the left piston **126** is illustrated in FIG. **6C**, for example. However, the positive pressure in the piston chamber **124** causes the valve flap **188** associated with the outlet channel **196** to deflect away from the valve seat **176** and flex about the valve post **212** to thereby allow the pressurized chemical product in the piston chamber **124** to flow into the outlet channel **196** and to the outlet **78** of the pump module **90** via the outlet tubing **216**. This valve configuration for the left piston **126** is illustrated in FIG. **7D**, for example.

[0105] Turning now to the right piston, during the negative pressure stroke of this piston, the negative pressure in the piston chamber **124** causes the valve flap **188** associated with the inlet channel **190** to deflect away from the valve seat **208** and flex about the valve post **180** to thereby allow the product in the inlet channel **190**, which is received from the inlet **76** of the pump module **90** via the inlet tubing **214**, to flow into the piston chamber **124**. This valve configuration for the right piston **126** is illustrated in FIG. **6D**, for example. However, the negative pressure in the piston chamber **124** causes the valve flap **188** associated with the outlet channel **196** to engage against the

annular valve seat **176** such that the valve is closed and fluid cannot pass from the piston chamber **124** to the outlet channel **196**. This valve configuration for the right piston **126** is illustrated in FIG. 7C, for example.

[0106] The left piston **126** continues to eject chemical product from the piston chamber **124** to the outlet channel **196**, and the right piston continues to pull chemical product into the piston chamber **124** from the inlet channel **190** until the left and right pistons **126** substantially reach their top dead position and bottom dead position, respectively. This configuration of the pump module **90** is shown in FIGS. 6B and 7B. At this point, the pistons **126** change direction with further activation of the motor **146** such that the left piston **126** begins to move downward through a negative pressure stroke and the right piston **126** begins to move upward through a positive pressure stroke.

[0107] For the left piston, during the negative pressure stroke of this piston, the negative pressure in the piston chamber **124** causes the valve flap **188** associated with the inlet channel **190** to deflect away from the valve seat **208** and flex about the valve post **180** to thereby allow the product in the inlet channel **190** to flow into the piston chamber **124**. This valve configuration for the left piston **126** is illustrated in FIG. 6D, for example. However, the negative pressure in the piston chamber **124** causes the valve flap **188** associated with the outlet channel **196** to engage against the annular valve seat **176** such that the valve is closed and fluid cannot pass from the piston chamber **124** to the outlet channel **196**. This valve configuration for the right piston **126** is illustrated in FIG. 7C, for example.

[0108] For the right piston, during the positive pressure stroke of this piston, the positive pressure in the piston chamber **124** causes the valve flap **188** associated with the inlet channel **190** to engage against the annular valve seat **208** such that the valve is closed and fluid cannot pass from the piston chamber **124** to the inlet channel **190**. This valve configuration for the left piston **126** is illustrated in FIG. 6C, for example. However, the positive pressure in the piston chamber **124** causes the valve flap **188** associated with the outlet channel **196** to deflect away from the valve seat **176** and flex about the valve post **212** to thereby allow the pressurized product in the piston chamber **124** to flow into the outlet channel **196**. This valve configuration for the right piston **126** is illustrated in FIG. 7D, for example.

[0109] The right piston **126** continues to eject product from the piston chamber **124** to the outlet channel **196**, and the right piston continues to pull product into the piston chamber **124** from the inlet channel **190** until the left and right pistons **126** substantially reach their bottom dead position and top dead position, respectively. This configuration of the pump module **90** is shown in FIGS. 6A and 7A. At this point, the pistons **126** change direction with further activation of the motor **146** such that the cycle described above repeats itself and product continues to be drawn into the pump module **90** and expelled from the pump module **90** in a substantially continuous and constant fashion.

[0110] The dual-piston arrangement of the pump module **90** provides a number of advantages. For example, it is believed that the valves **164** and the seals associated with the pistons **126** (e.g., the O-rings) will generally have a long operating life such that maintenance on the pump module **90** will be significantly reduced. By way of example, it is believed that the dual-piston pump module **90** may operate around 200% longer than current peristaltic pump designs. This is significant in both costs and down time for the chemical dispensing system. Additionally, the dual-piston arrangement provides a generally constant flow of chemical product from the pump during operation. This is in contrast to many types of pumps which may have generally non-continuous output cycles (e.g., step function output cycles). This may be important because of the amount of time in which to pump a chemical product to the washing machine may be relatively short. Because of the near constant flow of chemical product from the pump module **90**, a smaller pump may be utilized for achieving the desired amount of chemical product for delivery to the washing machine.

[0111] FIGS. 8-13B illustrate an improved pump module **240** in accordance with an embodiment of the invention. The pump module **240** is another type of module **66** used in the chemical dispenser

14 described above. In accordance with an aspect of the invention, the pump module **240** may be configured as a double-ended piston pump capable of relatively constant fluid flow over fairly short cycle times. The pump module **240** is also configured to be low maintenance and capable of very long run times before any maintenance operations are necessary to ensure the accurate dispensing of chemical product from the chemical dispenser **14**. This further reduces the maintenance costs and down time for the chemical dispensing system **10**.

[0112] To those and other ends, the exemplary pump module **240** of FIG. **8** in accordance with an embodiment of the invention is shown disassembled in FIG. **9**. As is shown and more specifically described below, the pump module **240** operates with a pumping action in a horizontal orientation rather than in a vertical orientation as is shown and described with reference, for example, to the pump module **90** shown in FIG. **5**. The pumping action, however, is not restricted to horizontal as all orientations of the piston are contemplated. While not shown, the pump module **240** includes a module housing, such as the module housing **92**, shown in FIG. **5**, which generally defines the cavity **118** to cover the internal components of the pump module **240**.

[0113] The internal components of the pump module **240** include a piston assembly **242**, a drive assembly **244**, and valve assembly **246**, **248**. While a front housing portion is not shown in FIG. **9**, the front housing portion **100** shown in FIG. **5** may be utilized in conjunction with a rear housing portion **250** which fit together to form the housing **92** with the interior **104** for housing the components of the piston assembly **242**. The rear housing portion **250** provides a generally planar wall **252** from which spindles **254** extend for mounting the piston assembly **242**. A drive aperture **256** is located in the wall **252** relative to the spindles **254** and receives the drive shaft **150** of the motor **146**. On the drive shaft **150**, a connecting shaft **260** is secured. The connecting shaft **260** is generally circular and receives the drive shaft **150** at its center. An eccentrically located pin **262** extends from a face of the connecting shaft **260** opposite the drive shaft **150**. Rotation of the drive shaft **150** rotates the connecting shaft **260** with the pin **262** tracing a circular path defined by the offset between the axis of the drive shaft **150** and the axis of the pin **262**. The circular path traced by the pin **262** energizes the piston assembly **242** as is further described below with reference to FIGS. **12-13B**.

[0114] With continued reference to FIGS. **9** and **12**, front and rear piston chamber housings **264**, **266** are assembled together and cooperate to form a piston chamber **270**. The piston chamber **270** may be symmetrically formed about a mid-plane of the housings **264**, **266**. The housings **264**, **266** define cylinder walls in the piston chamber **270** so as to form a left cylinder **272** opposing a right cylinder **274** separated by a yoke cavity **276** (labeled in FIG. **12**). Unlike the module **90**, for example, shown in FIG. **5**, the piston assembly **242** has only two piston cylinders **272** and **274** that lie along a common longitudinal axis. That is, the piston assembly **242** does not include more or have less than two cylinders. In the yoke cavity **276**, there is an opening **278** in the rear piston chamber housing **266** that receives the connecting shaft **260**. In this way, the pin **262** extends into the piston chamber **270** to mechanically drive a piston.

[0115] In the exemplary embodiment shown, and with reference to FIGS. **9** and **12**, portions of each cylinder **272** and **274** are defined by corresponding cylinder heads **280**, **282**. As can be appreciated from FIG. **9**, the cylinder heads **280**, **282** are received between the front and rear piston chamber housing **264** and **266**. Fastening the front housing **264** and rear housing **266** together via fasteners, such as by the screws shown, secures the cylinder heads **280**, **282** in a fixed position at each end of the piston chamber **270**. In the exemplary embodiment, the cylinder heads **280**, **282** together with the housing **264**, **266** define cylinders **272** and **274**.

[0116] Fluid flow is directed to and from the cylinder **272** and **274**, as is described below, via valve assemblies **246**, **248**, which are coupled to corresponding cylinder heads **280**, **282**. As shown, each valve assembly **246**, **248** includes an inlet valve housing **246a**, **248a** and an outlet valve housing **246b**, **248b**. Inlet tubing **286** and outlet tubing **290** are connected to respective valve assemblies **246**, **248** for directing fluid to/from the piston assembly **242**. A plurality of valves **284** are captured

between housings **246a**, **246b**, **248a**, and **248b** corresponding cylinder head **280**, **282**. Each valve **284** controls fluid flow in a predetermined direction during operation of the piston assembly **242**. In the exemplary embodiment shown, the valves **284** are duckbill valves. However, embodiments of the invention are not limited to duckbill valves, as other one-way fluid flow valves may be utilized in accordance with embodiments of the invention.

[0117] With reference to FIGS. **9**, **10**, and **11**, a piston **292** is movably received between housings **264** and **266** in the piston chamber **270**. The piston **292** is shown best in FIGS. **10** and **11** and has a left piston head **292a** and a right piston head **292b**, which are received in the left and right cylinders **272**, **274**, respectively. The piston **292** is referred to as a double-ended piston because it has two working heads. As described, a single cycle of the piston **292** produces two chemical product exhausts from the module **240** and two chemical product intakes into the module **240**. The piston head **292a** and the piston head **292b** extend from a sliding yoke **294**, which is movably received in the yoke cavity **276**. The yoke cavity **276** is larger in the horizontal direction than the corresponding width of the sliding yoke **294** but is only slightly larger than the sliding yoke **294** in the vertical or height direction. With these relative dimensions, the piston **292** is capable of moving side-to-side. In the exemplary embodiment shown, the piston head **292a** and the piston head **292b** lie on a longitudinal axis **288**. The piston may be symmetrical about a plane that intersects the longitudinal axis **288** and about a plane that divides the over length in half. An elliptical slot **296** in the sliding yoke **294** receives the pin **262** of the drive assembly **244** when the piston **292** is contained in the piston chamber **270**.

[0118] Rotation of the pin **262** about a center of the connecting shaft **260** causes the pin **262** to frictionally engage the elliptical slot **296** as the pin **262** rotates along a path defined by its eccentricity. This eccentric rotation of the pin **262** is transmitted to the piston **292**, which reciprocates along a linear path, i.e., in a side-to-side motion by a distance determined by the eccentricity of the pin **262**. In the exemplary embodiment, that motion is horizontal relative to the vertical movement of the pistons **126** in embodiment of the pump module **90** shown in FIG. **5**, for example. Lower and upper slide rails **300**, **302** of the sliding yoke **294** may contact and slide in cooperation with adjacent surfaces of the yoke cavity **276** to guide the side-to-side movement of the piston **292** in the cavity **276**. Further in that regard, bearings **304** (shown in FIGS. **9** and **12**) may slidably engage piston heads **292a** and **292b** and may further guide reciprocating motion of the piston **292** in the piston chamber **270** during fluid pumping, described below. By way of example only, and not limitation, bearings **304** may be scarf bearings.

[0119] As shown in FIGS. **11** and **12**, the piston heads **292a** and **292b** may be hollow and open to the corresponding cylinder **272**, **274**. Advantageously, when the piston **292** is formed as a single-piece molded component, such as from a plastic, the piston heads **292a** and **292b** may include hollow end portions **298a** and **298b**. This design permits the surface engagement with the cylinders **272** and **274** to be of more precise dimensional tolerance and reduces gaps in the fit between the piston **292** and the cylinders **272**, **274**. Fluid leakage is thereby reduced while pumping efficiency/accuracy of the pump module **240** is improved. In the exemplary embodiment, head seals **306** are captured between the housings **264** and **266** and a respective one of the cylinder heads **280**, **282** to fluidly seal the cylinders **272**, **274** from fluid leakage between piston head **292a**, **292b**; cylinder heads **280**, **282**; and housings **264**, **266**.

[0120] Operation of the pump module **240**, once coupled to the chemical dispenser **14** and operational within the chemical dispensing system **10**, will now be described with reference to FIGS. **12**, **12A**, and **12B** during one lateral motion of the piston **292** and with reference to FIGS. **13**, **13A**, and **13B** during the opposite lateral motion of the piston **292**. The two lateral motions are one full cycle of the piston **292**. To that end, rotation of the connecting shaft **260** causes the pin **262** to also rotate clockwise, and by its engagement with the sliding yoke **294** at the elliptical slot **296**, strokes the piston **292** to the left. While a clockwise rotation of the pin **262** is described, counterclockwise rotation is also contemplated and embodiments of the invention are not limited to

either clockwise rotation or counterclockwise rotation.

[0121] Clockwise rotation is illustrated in FIG. 12 by arrow 310 for rotation of the pin 262. The clockwise rotation of pin 262 causes the piston 292 to move according to arrows 312 in each of FIGS. 12, 12A, and 12B. With reference to FIG. 12A, lateral motion of the piston 292 (and piston head 292a) pushes chemical product in the left cylinder 272 out of the cylinder head 280 and through the valve 284 at the outlet valve housing 246b. The valve 284 at this location is a one-way valve that opens to allow fluid flow in the direction of arrows 316. The valve 284 in the inlet valve housing 246a is closed and prevents fluid from exiting the cylinder 272 at this location as the piston 292 moves laterally to the left.

[0122] Simultaneously, as the piston 292 strokes laterally to the left and exhausts chemical product from the left cylinder 272, and with reference to FIG. 12B, chemical product is pulled into the right cylinder 274 according to arrow 322 through the valve 284 at the inlet valve housing 248a. The valve 284 in the inlet valve housing 248a is a one-way valve that opens to allow fluid flow in the direction of arrows 322. The valve 284 in the outlet valve housing 248b is closed and prevents fluid from entering the right cylinder 274 at this location as the piston 292 moves laterally to the left. With reference to FIGS. 12A and 12B, chemical product flows out of the left cylinder 272 toward the washing machine 12a (FIG. 1), for example, while fluid is drawn into the right cylinder 274 from one of the chemical reservoirs 18a, 18b, for example.

[0123] Continued rotation of the pin 262 in a clockwise direction from the position shown in FIG. 12 continues fluid pumping, but from the right cylinder 274 to the washing machine 12a. In that regard, rotation of the connecting shaft 260 further clockwise from the position shown in FIG. 12 to the position shown in FIG. 13 requires that the pin 262 also rotates clockwise. By its engagement with the sliding yoke 294 at the elliptical slot 296, the piston 292 is moved laterally to the right in the piston chamber 270. The clockwise rotation of pin 262 causes the piston 292 to move according to arrows 326 in each of FIGS. 13, 13A, and 13B.

[0124] Fluid motion in the left cylinder 272 is described with reference to FIGS. 13 and 13A. Lateral motion of the piston 292 and piston head 292a in the left cylinder 272, pulls fluid into the left cylinder 272 according to arrow 332 through the valve 284 at the inlet valve housing 246a. The valve 284 in the inlet valve housing 246a is a one-way valve that opens to allow fluid flow in the direction of arrows 332. The valve 248 in the outlet valve housing 246b is closed and prevents fluid from entering the left cylinder 272 at this location as the piston 292 moves laterally to the right. In this way, fluid fills the left cylinder 272.

[0125] Simultaneously, as the piston 292 strokes laterally to the right, it exhausts chemical product from the right cylinder 274 and consequently out of the pump module 240. The fluid exits the right cylinder 274 out of the cylinder head 282 and through the valve 284 at the outlet valve housing 248b. The valve 284 at this location is a one-way valve that opens to allow fluid flow in the direction of arrows 330. The valve 284 in the inlet valve housing 248a is closed and prevents fluid from exiting the cylinder 274 at this location as the piston 292 moves laterally to the right. With the rotation of the connecting shaft 260, the piston 292 is moved from side-to-side. At one cylinder 272, 274, fluid is expelled from the pump module 240 to downstream equipment, such as the washing machine. At the same time, at the opposite cylinder 272, 274, fluid is drawn in. With this double-ended piston, a single piston cycles provides both chemical product inflow and outflow at each end during one 360° rotation of the drive assembly 244.

[0126] The double-ended piston 292 of the pump module 240 is advantageous. For example, it is believed that the valves 284 will generally have a long operating life such that maintenance on the pump module 240 will be significantly reduced. By way of example, it is believed that the double-ended piston pump module 240 may operate around 200% longer than current peristaltic pump designs due to a reduction in the number of moving parts. This is significant in both costs and down time for the chemical dispensing system. Additionally, the double-ended arrangement provides a generally constant flow of chemical product from the pump during operation. The back

and forth motion of the piston **292** produces a nearly continuous supply of fluid downstream. Advantageously, because of the double-ended piston design, the timing of fluid motion from left side and right side is constant. There is no need to consider the relative position of each separate piston as in a two separate piston pump. In the embodiment shown in FIG. **9**, the timing of the pumping action is fixed at 180 degrees. Moreover, the volume of fluid expelled from the left and right sides is equal.

[0127] This is in contrast to many types of pumps which may have generally non-continuous output cycles (e.g., step function output cycles). This may be important because of the amount of time in which to pump a chemical product to the washing machine may be relatively short. Because of the near constant flow of chemical product from the pump module **240**, a smaller pump may be utilized for achieving the desired amount of chemical product for delivery to the washing machine. [0128] FIGS. **14-21B** illustrate an improved pump module **340** in accordance with an embodiment of the invention. The pump module **340** is one of the types of modules **66** used in chemical dispenser **14** described above. In accordance with an aspect of the invention, the pump module **340** may be configured as a dual-piston pump that is capable of relatively constant fluid flow over fairly short cycle times. The dual-piston pump module **340** is similar in some respects to the dual-piston pump module **90**, described above, and is also configured to be low maintenance and capable of very long run times before any maintenance operations are necessary to ensure the accurate dispensing of chemical product from the chemical dispenser **14**. This further reduces the maintenance costs and down time for the chemical dispensing system **10**.

[0129] A disassembled dual-piston pump module **340** in accordance with an embodiment of the invention is illustrated in FIG. **15**. The dual-piston pump module **340** includes a piston assembly **342**, a drive assembly **344**, and a valve assembly **346**. Although not shown in FIG. **15**, the module housing **92** described with the pump module **90** and shown in FIG. **5** may be utilized to house the pump module **340**. The rear housing portion **350** includes a generally planar wall **352**, a generally U-shaped support or frame **354** extending from an inner surface of the wall **352**, a pair of spindles **358** extend from the wall **352** within the U-shaped frame **354**, and a trio of support posts **360** extend from the wall **352** outboard of the U-shaped frame **354**. The rear housing portion **350** further includes a drive aperture **362** in the wall **352** centrally located above and between the spindles **358** and a pair of slots **364**, the purpose of which is described above with regard to the pump module **90**, at a lower end of the rear housing portion **350**. Although not shown, a front housing portion similar to that shown in FIG. **5** generally defines a cavity and effectively operates as a cover for the internal components of the pump module **340**.

[0130] As illustrated in FIGS. **14** and **15**, the piston assembly **342** includes a piston chamber housing **370** secured to the rear housing portion **350**. The piston chamber housing **370** defines a pair of piston cavities **380**, **382** that movably receive pistons, described below. In the exemplary embodiment shown, the piston chamber housing **370** is composed of two separate half housings **374** and **376** that are secured together via screws or by other means. Each half housing **374** and **376** define cylinder cavities **380** and **382** so that when assembled together, the cylinder cavities **380** and **382** collectively define the right and left piston chambers **372a**, **372b**. In that regard, the piston chambers **372a**, **372b** define two pairs of upper and lower cylinders **384a**, **384b** and **388a**, **388b** and left and right yoke cavities **390a** and **390b**. Collectively, left cylinders **384a**, **384b** and left yoke cavity **390a** movably receive one piston and, similarly, right cylinder **388a**, **388b** and right yoke cavity **390b** movably receives the other piston.

[0131] In the exemplary embodiment, the piston chamber housing **370** includes a pair of cylinder heads **392a**, **392b** that are captured between the separate half housings **374** and **376**. The cylinder heads **392a**, **392b** include cylinder walls **394** that align with the cylinder cavities **380** and **382** and so may form an end portion of each respective cylinder **384a** and **388a**. Only one set of cylinders **384a** and **388a** (i.e., the upper cylinders) may be formed with cylinder heads **392a**, **392b**. Head seals **306** (described with reference to FIG. **9**) are captured between the housings **374** and **376** and

a respective one of the cylinder heads **392a**, **392b** to fluidly seal the cylinders **384a**, **388a** from fluid leakage. Although not shown, the cylinder heads **392a**, **392b** may fully form one or both the left and right cylinders **384a**, **388a**. The set of cylinders **384b** and **388b** opposing the cylinders **384a** and **388a** may be closed off by the piston chamber housing **370** at **378** (shown best in FIG. **16A**) to form a blind bore at that location. Because the cylinders **384a**, **388a** are closed off, no fluid enters or exits this portion of the piston chambers **372a**, **372b**.

[0132] With reference to FIGS. **15**, **15B**, and **15C**, a pair of pistons **396**, **398** is movably received within a respective piston chambers **372a**, **372b** of the piston chamber housing **370**. In the exemplary embodiment shown, each piston **396**, **398** is double ended. That is, each piston **396**, **398** includes two end portions or heads **400a**, **400b** and **402a**, **402b** that extend from a sliding yoke **404**, **406**, respectively, and so are similar to the double-ended piston shown in FIG. **9**, for example. However, by contrast, while having two working ends, the pistons **396**, **398** pump chemical product at one end, not both. In the exemplary embodiment shown, the piston head **400a** and the piston head **400b** lie on a longitudinal axis **414**. The piston head **402a** and the piston head **402b** also share a separate, common longitudinal axis **414**. As shown in FIGS. **15C** and **16A**, the piston heads **400a**, **400b**, **402a**, **402b** are hollow and open to the corresponding cylinder **384a**, **388a**. This design is advantageous for at least the same reasons identified above with reference to piston **292** shown in FIG. **9**. As is shown best in FIG. **15C**, each of the heads **400b** and **402b** may be shorter in length as measured from the corresponding sliding yoke **404**, **406** than the opposing heads **400a** and **402a**.

[0133] The sliding yokes **404**, **406** are each a generally rectangular portion of the piston **396**, **398** and may have opposed slide rails **410** and **412**. Each sliding yoke **404**, **406** is movably received in a respective yoke cavity **390a**, **390b** such that the slide rails **410** and **412** frictionally engage corresponding slide surfaces of the yoke cavities **390a**, **390b** during movement of the piston **396**, **398**. This sliding engagement facilitates guided, reciprocating motion of the piston **396**, **398** in the respective cavities **380**, **382**. Additionally, guided engagement is produced between the cylinders **384b**, **388b** and a respective one of the pistons **396**, **398**. While the cylinders **384b**, **388b** do not participate in movement of fluid because they are each blind bores, engagement between the cylinders **384b**, **388b** and the piston heads **400b** and **402b** provides additional alignment to the reciprocating motion. Thus, the double-ended feature of the piston **396**, **398** improves alignment between the piston **396**, **398** in the opposing cylinders **384a**, **388a**. This is advantageous because it improves pumping efficiency and reduces wear. Longevity of the piston assembly **342** is thus increased. To further aid in guiding reciprocating movement of each left cylinder **384a** and **384b** and each right cylinder **388a** and **388b** includes a bearing **304**, described above with reference to FIG. **9**. Each sliding yoke **404**, **406** includes an elliptical slot **416** which receives one pin **156** of the drive assembly **344** through the piston chamber housing **370**, shown in FIG. **15**.

[0134] As illustrated in FIG. **15**, the drive assembly **344** may be substantially identical to the drive assembly **96** described above with reference to FIG. **5**. When the motor **146** is energized, the primary drive gear **152** drives the secondary drive gears **154**, which in turn cause reciprocating movement of the pistons **396**, **398** within their respective piston chambers **372a**, **372b**. The use of a dual-piston, double-ended arrangement as a pump, however, involves the coordinated use of a valve arrangement, to which we now turn.

[0135] As shown in FIG. **15**, the valve assembly **346** is coupled to the cylinder heads **392a**, **392b**. The valve assembly **346** includes a valve housing **420**, two pairs of valves **422**, and a product manifold **424**. The valve assembly **346** controls fluid flow into and out of the piston assembly **342**. As illustrated in more detail in FIG. **15A**, the valve housing **420** includes two pair of fluid ports **426a**, **426b** and **428a**, **428b** each of which is in fluid communication with a respective one of the valves **422** and a respective one of the cylinders **384a**, **388a** via one of the cylinder heads **392a**, **392b**. The product manifold **424** includes matching fluid ports **430a** and **430b** and **432a** and **432b**. The valves **422** are oriented such that one valve permits fluid to enter the cylinder **384a**, **388a** and one valve permits fluid to exit the cylinder **384a**, **388a**. Similar to the valves **284**, shown in FIG. **9**,

in an exemplary embodiment, the valves **422** are duckbill valves or other one-way flow control valves. In that regard, each valve **422** is seated in a valve housing **434a**, **434b** and **436a**, **436b**. As shown, valve housings **434a** and **436a** extend from a planar support plate **440**. And, valve housings **434b** and **436b** extend from the product manifold **424**. The lateral ends of the planar support plate **440** include a support posts **442** which are received in matching bores **444** in the product manifold **424**. When the product manifold **424** and the support plate **440** are assembled, the valves **422** are secured in their respective housings **434a**, **434b**, **436a**, **436b**.

[0136] With continued reference to FIG. **15A**, the product manifold **424** provides for chemical product flow to and from the valve assembly **346** and the piston assembly **342**. To that end, the product manifold **424** includes an inlet channel **446** having a connector **450** at one end and is closed off at the other end **452** and an outlet channel **448** having a connector **454** at one end and is closed off at the other end **456**. The product manifold **424** is configured to be coupled to the cylinder heads **392a**, **392b**, as described above. The fluid ports **430b** and **432b** are configured to be in selective communication with the inlet channel **446**, and the outlet ports **430a** and **432a** are configured to be in fluid communication with the outlet channel **448**. As further shown in FIGS. **15**, **16A**, **16B**, **17A**, and **17B**, the valve assembly **346** further includes inlet and outlet tubing **214**, **216** extending from their respective connectors **450**, **454** to connectors **218** of the pump module **340**.

[0137] As is illustrated in FIGS. **16A**, **17A**, and **17B**, when the piston assembly **342**, the valve assembly **346**, and the product manifold **424** are coupled together, the outlet channel **448** is in fluid communication with each of the left cylinder **384a** and the right cylinder **388a** through a respective one of the valves **422**. Thus, each cylinder **384a** and **388a** has associated therewith an outlet port **426a**, **428a** and **430a**, **432a** for allowing chemical product out of the cylinder **384a**, **388a** and into the outlet channel **448**.

[0138] And, with reference to FIGS. **16B**, **18A**, and **18B**, when assembled, the inlet channel **446** is in fluid communication with each of the left cylinder **384a** and the right cylinder **388a** through a respective one of the valves **422**. Thus, each cylinder **384a** and **388a** has associated therewith an inlet port **426b**, **428b** and **430b**, **432b** for allowing intake of the chemical product into the respective cylinder **384a** and **388a** from the inlet channel **446**.

[0139] Operation of the pump module **340**, once coupled to the chemical dispenser **14** and operational within the chemical dispensing system **10**, will now be described. FIGS. **16A-21B** illustrate operation of the pump module **340** as it relates to inflow and outflow of chemical product from the pump module **340**. The initial configuration described of the pump module **340** will be with the left piston **396** in the bottom dead position with the cylinder **384a** full of product, and the right piston **398** in the top dead position with the cylinder **388a** discharged. This configuration is shown in FIGS. **16A** and **16B** with respect to the inlet channel **446** and outlet channel **448**, respectively, of the product manifold **424**. Although not shown, it will be appreciated that following discharge, each cylinder **384a**, **388a** may contain residual product, particularly in the hollow heads **400a**, **402a**. Activation of the motor **146** (i.e., under the control of controller **34**) causes the primary drive gear **152** to rotate, which in turn causes both the secondary drive gears **154** to rotate (as is indicated by arrows **460**). With rotation of the secondary drive gears **154**, the left piston **396** begins to move upward (indicated by arrow **462**) through a positive pressure stroke (i.e., exhaust) and the right piston **398** begins to move downward (indicated by arrow **464**) through a negative pressure stroke (i.e., intake or vacuum).

[0140] Focusing first on outlet channel **448**, during the positive pressure stroke of the left piston **396** (shown by way of arrow **462**), the positive pressure in the cylinder **384a** causes the valve **422** between the fluid ports **426a** and **430a** to open. When opened, fluid in the cylinder **384a** is permitted to flow into the outlet channel **448**. This valve configuration for the left piston **396** is illustrated in FIG. **17A**, for example. However, during the negative pressure stroke of the piston **398**, the negative pressure in the cylinder **388a** causes the valve **422** between the fluid ports **428b** and **432b** to remain closed. This is shown in FIGS. **16A** and **17B**, for example. When that valve **422**

is closed, fluid is prevented from passing from the outlet channel **448** into the cylinder **388a**. [0141] Turning now to the inlet channel **446** and FIG. **16B**, during the same positive pressure stroke of the left piston **396** (described above and shown by way of arrow **462**), the positive pressure in the cylinder **384a** causes the valve **422** between the fluid ports **426b** and **430b** to remain closed. When closed, fluid in the cylinder **384a** is prevented from flowing into the inlet channel **446**. This valve configuration for the left piston **396** at the inlet channel **446** is illustrated in FIG. **18A**, for example. However, during the negative pressure stroke of the piston **398**, the negative pressure in the cylinder **388a** causes the valve **422** between the fluid ports **428b** and **432b** to open. This is shown in FIGS. **16B** and **18B**, for example. When that valve **422** is opened, fluid is drawn into the cylinder **388a** from the inlet channel **446**.

[0142] The left piston **396** continues to exhaust chemical product from the cylinder **384a** to the outlet channel **448** (shown in FIGS. **16A** and **17A**), and the right piston **398** continues to pull chemical product into the cylinder **388a** from the inlet channel **446** (shown in FIGS. **16B** and **18B**) until the left piston **396** and right piston **398** substantially reach their top dead position and bottom dead position, respectively. This configuration of the pump module **340** is shown in FIGS. **19A** and **19B**. At this point, the pistons **396**, **398** change direction with further activation of the motor **146** such that the left piston **396** begins to move downward through a negative pressure stroke and the right piston **398** begins to move upward through a positive pressure stroke.

[0143] At this position and referring to FIGS. **19A** and **20A**, which depict a cross section through the outlet channel **448**, during the negative pressure stroke of the left piston **396** (shown by way of arrow **464**), the negative pressure in the cylinder **384a** causes the valve **422** between the fluid ports **426a** and **430a** to remain closed. When that valve **422** is closed, fluid is prevented from passing from the outlet channel **448** to the cylinder **384a**. This valve configuration for the right piston **398** is illustrated in FIG. **20A**, for example. However, with reference to FIGS. **19A** and **20B**, the positive pressure in the right cylinder **388a** causes the valve **422** between the fluid ports **428b** and **432b** to open. When that valve **422** is opened, fluid in the cylinder **388a** is permitted to flow into the outlet channel **448**. This valve configuration for the right piston **398** is illustrated in FIG. **20B**, for example.

[0144] Turning now to the inlet channel **446** and referring to FIGS. **19B** and **21B**, during the positive pressure stroke of the right piston **398** (shown by way of arrow **462**), the positive pressure in the cylinder **388a** causes the valve **422** between the fluid ports **428a** and **432a** to remain closed. When that valve **422** is closed, fluid is prevented from flowing from the cylinder **388a** to the inlet channel **446**. This valve configuration for the left piston **396** is illustrated in FIG. **21B**, for example. However, with reference to FIGS. **18B** and **21A**, the negative pressure in the left cylinder **384a** causes the valve **422** between the fluid ports **426b** and **430b** to open. When that valve **422** is opened, fluid in the inlet channel **446** is permitted to flow into the cylinder **384a**. This valve configuration for the left piston **396** is illustrated in FIG. **21A**, for example.

[0145] With reference to FIGS. **19A** and **19B**, the right piston **398** continues to eject product from the cylinder **388a** to the outlet channel **448**, and the left piston **396** continues to intake product into the cylinder **384a** from the inlet channel **446** until the left and right pistons **396**, **398** substantially reach their bottom dead position and top dead position, respectively. This configuration of the pump module **340** is shown in FIGS. **16A** and **16B**. At this point, the pistons **396**, **398** change direction with further activation of the motor **146** such that the cycle described above repeats itself and product continues to be drawn into the pump module **340** and expelled from the pump module **340** in a substantially continuous and constant fashion.

[0146] The dual-piston double-ended arrangement of the pump module **340** provides a number of advantages. For example, it is believed that the valves **422** and the seals (e.g., the O-rings) associated with the pistons **396**, **398** will generally have a long operating life such that maintenance on the pump module **340** will be significantly reduced. By way of example, it is believed that the dual-piston pump module **340** may operate around 200% longer than current peristaltic pump

designs. This is significant in both costs and down time for the chemical dispensing system. Additionally, the dual-piston arrangement provides a generally constant flow of chemical product from the pump during operation. This is in contrast to many types of pumps which may have generally non-continuous output cycles (e.g., step function output cycles). This may be important because of the amount of time in which to pump a chemical product to the washing machine may be relatively short. Because of the near constant flow of chemical product from the pump module 340, a smaller pump may be utilized for achieving the desired amount of chemical product for delivery to the washing machine.

[0147] While the present invention has been illustrated by a description of various preferred embodiments and while these embodiments have been described in some detail, it is not the intention of the Applicant to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. The various features of the invention may be used alone or in numerous combinations depending on the needs and preferences of the user.

Claims

1.-13. (canceled)

14. A pump module for a modular chemical dispensing system, comprising: a module housing; a piston assembly; a drive assembly, and a valve assembly.

15. The pump module of claim 14, wherein the piston assembly comprises: a piston housing defining at least two piston cylinders; at least two pistons each defining a base and a piston head for positioning in respective piston cylinders, wherein the base of the at least two pistons are operatively coupled to the drive assembly for reciprocating the pistons relative to the piston cylinders.

16. The pump module of claim 15, wherein the piston housing includes at least one guide channel, and each piston including at least one guide rod, wherein the at least one guide rod is configured to be received in a respective guide channel for guiding movement of the pistons.

17. The pump module of claim 15, wherein the drive assembly comprises: a motor having a drive shaft coupled to the module housing; and a gear arrangement operatively coupled to the motor and further operatively coupled to the piston assembly.

18. The pump module of claim 17, wherein the gear arrangement comprises: a primary drive gear coupled to the drive shaft of the motor; and a pair of secondary drive gears configured to be driven by the primary drive gear.

19. The pump module of claim 18, wherein each of the secondary drive gears includes a pin eccentrically positioned relative to a rotational axis of the secondary drive gears, and wherein each pin is configured to be received within a slot in the base of the pistons for moving the pistons.

20. The pump module of claim 14, wherein the valve assembly comprises: a valve housing; a pair of valves; and a product manifold.

21. The pump module of claim 20, wherein the valve housing comprises: a pair of valve heads, wherein each valve head includes a valve recess, wherein each valve recess includes an inlet port, an outlet port, and a valve seat, wherein the valve seat is configured to receive one of the pair of valves, and wherein the inlet and outlet ports of each valve head are in communication with a respective piston cylinder of the piston assembly.

22. The pump module of claim 21, wherein the inlet port includes at least one flow aperture and a valve post.

23. The pump module of claim 22, wherein the inlet port includes a pair of flow apertures with the valve post disposed therebetween.

24. The pump module of claim 21, wherein the outlet port includes an annular valve seat.

25. The pump module of claim 21, wherein the product manifold comprises: an inlet channel; and

an outlet channel, wherein the inlet channel is in communication with the inlet port of each of the valve heads, and wherein the outlet channel is in communication with the outlet port of each of the valve heads.

26. The pump module of claim 14, wherein the piston assembly comprises: a piston housing defining at least two piston cylinders; and a piston having a sliding yoke and two piston heads extending in opposing directions from the sliding yoke, each one of the piston heads being received in a respective piston cylinder, wherein the sliding yoke is operatively coupled to the drive assembly for reciprocating the piston relative to the two piston cylinders.

27. The pump module of claim 26, wherein one cycle of the piston in the piston assembly is configured to produce two exhaust and two intake cycles.

28. The pump module of claim 26, wherein the piston heads share a common longitudinal axis.

29. The pump module of claim 26, wherein the two piston heads are of different lengths.

30. The pump module of claim 26, wherein the piston heads are hollow and are open to the respective piston cylinder.

31. The pump module of claim 26, wherein the sliding yoke defines an elliptical slot and the drive assembly is movably coupled to the sliding yoke by the elliptical slot.

32. The pump module of claim 26, wherein the piston housing further defines an opening in a yoke cavity, the yoke cavity receives the sliding yoke and the drive assembly engages the piston assembly through the opening.

33. The pump module of claim 26, wherein the piston assembly further comprises: a first cylinder head secured in the piston housing and at least partially defining a portion of one piston cylinder; and a second cylinder head secured in the piston housing and at least partially defining a portion of the other piston cylinder.

34. The pump module of claim 33, wherein each of the first cylinder head and the second cylinder head are in fluid communication with one piston.

35. The pump module of claim 33, wherein the first cylinder head is in fluid communication with a first piston and the second cylinder head is in fluid communication with a second piston, the first piston and the second piston are different pistons.

36. The pump module of claim 26, wherein the valve assembly comprises: an inlet valve housing including an inlet valve in fluid communication with at least one cylinder; and an outlet valve housing including an outlet valve in fluid communication with the at least one cylinder.

37. The pump module of claim 36, wherein each valve is a duckbill valve.

38. The pump module of claim 26, wherein the piston housing defines two cylinders and the piston assembly further comprises: a first cylinder head secured in the piston housing and at least partially defining a portion of one piston cylinder; a second cylinder head secured in the piston housing and at least partially defining a portion of the other piston cylinder; and the valve assembly comprising: a first inlet valve housing including a first inlet valve coupled to the first cylinder head and a first outlet valve housing including a first outlet valve coupled to the first cylinder head, and a second inlet valve housing including a second inlet valve coupled to the second cylinder head and a second outlet valve housing including a second outlet valve coupled to the second cylinder head, wherein each of first inlet valve, the first outlet valve, the second inlet valve, and the second outlet valve is a duckbill valve.

39. A pump module for a modular chemical dispensing system, comprising: a module housing; a piston assembly comprising: a piston housing defining at least two piston cylinders; at least two pistons each defining a base and a piston head for positioning in respective piston cylinders a drive assembly comprising: a motor having a drive shaft coupled to the module housing; and a gear arrangement operatively coupled to the motor and further operatively coupled to the piston assembly; and a valve assembly comprising: a valve housing comprising a pair of valve heads; a pair of valves; and a product manifold; wherein each valve head includes a valve recess, wherein each valve recess includes an inlet port, an outlet port, and a valve seat, wherein the valve seat is

configured to receive one of the pair of valves, wherein the inlet and outlet ports of each valve head are in communication with a respective one of the piston cylinders, and wherein the base of the at least two pistons are operatively coupled to the drive assembly for reciprocating the pistons relative to the piston cylinders.
